Dummy Variables and Interaction

1 Qualitative regressors

Qualitative regressors

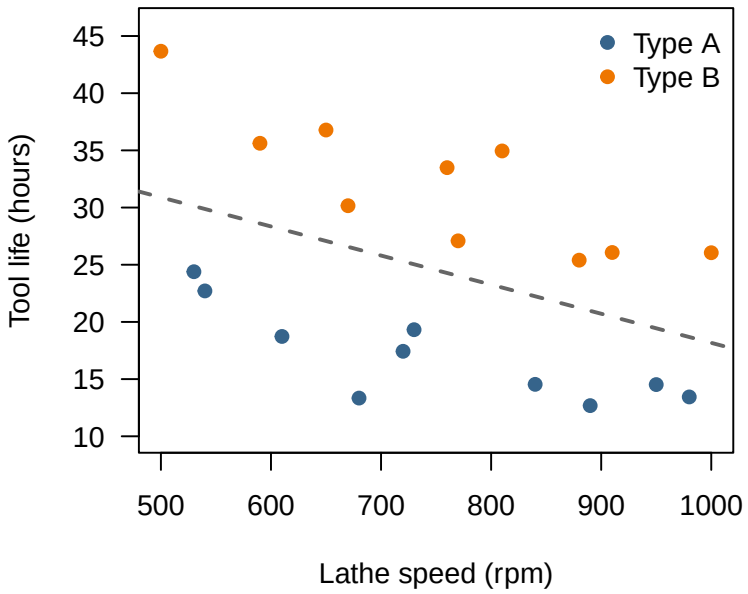
- So far every regressor has been quantitative: speed, temperature, delivery volume.
- Many important predictors are qualitative (categorical): sex, region, treatment group, machine type.
- A category has no numerical value, so it cannot enter $\mathbf{X}$ as it is.
- **Question:** how can we put "tool type A or B" into a regression model?

Example dataset: tool life data

- y: hours of tool life; x_1: lathe speed (rpm); two tool types, A and B, $n = 20$.

Tool type A			Tool type B		
i	y	x_1	i	y	x_1
1	18.73	610	11	30.16	670
2	14.52	950	12	27.09	770
3	17.43	720	13	25.40	880
4	14.54	840	14	26.05	1000
5	13.44	980	15	33.49	760
6	24.39	530	16	35.62	590
7	13.34	680	17	26.07	910
8	22.71	540	18	36.78	650
9	12.68	890	19	34.95	810
10	19.32	730	20	43.67	500

What happens if we ignore tool type?



- Simple linear regression of y on x_1 (dashed): $\hat{y} = 43.62 - 0.0254\, x_1$.
- $R^2 = 0.186$, $\hat{\sigma} = 8.44$.
- The points split into two clusters: type B tools last longer at every speed.
- Tool type is a missing regressor; the residuals carry its effect.

2 Indicator variables

Indicator (dummy) variable

- An indicator variable encodes membership in a category with the values 0 and 1.
- For tool type:

$$x_2 = \begin{cases} 0 & \text{if the observation is from tool type A,} \\ 1 & \text{if the observation is from tool type B.} \end{cases}$$

- Type A is coded 0: it is the **reference** (baseline) category.

Tool life data with the indicator column

Tool type A				Tool type B			
i	y	x_1	x_2	i	y	x_1	x_2
1	18.73	610	0	11	30.16	670	1
2	14.52	950	0	12	27.09	770	1
3	17.43	720	0	13	25.40	880	1
4	14.54	840	0	14	26.05	1000	1
5	13.44	980	0	15	33.49	760	1
6	24.39	530	0	16	35.62	590	1
7	13.34	680	0	17	26.07	910	1
8	22.71	540	0	18	36.78	650	1
9	12.68	890	0	19	34.95	810	1
10	19.32	730	0	20	43.67	500	1

$x_2 = 0$ for tool type A and $x_2 = 1$ for tool type B; x_2 is a new column of $\mathbf{X}$ next to $\mathbf{1}$ and x_1.

Model with one quantitative regressor and one indicator

- Regress y on speed x_1 and the indicator x_2:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

- This is an ordinary multiple linear regression; $\mathbf{X}$ has columns $\mathbf{1}$, $\mathbf{x}_1$, $\mathbf{x}_2$, and OLS applies unchanged.
- Only the interpretation of the coefficient on x_2 is new.

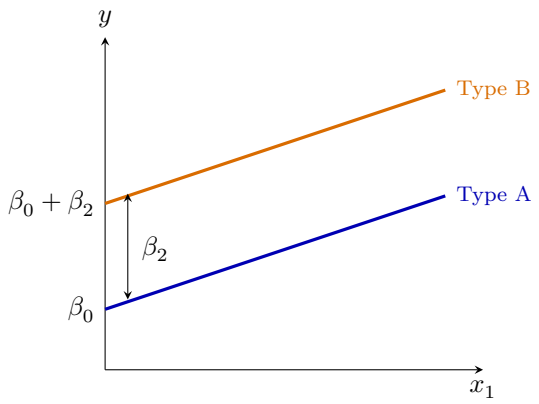
Two regression lines from one model

- Setting $x_2 = 0$ and $x_2 = 1$ gives the mean response for each tool type:

$$\begin{aligned} \text{Type A } (x_2 = 0): &\quad \mathbb{E}(y \mid x_1) = \beta_0 + \beta_1 x_1, \\ \text{Type B } (x_2 = 1): &\quad \mathbb{E}(y \mid x_1) = (\beta_0 + \beta_2) + \beta_1 x_1. \end{aligned}$$

- The lines have the same slope β_1 and intercepts that differ by β_2: the lines are **parallel**.
- β_2 is the difference in mean tool life between type B and type A at any fixed speed.

Geometry: parallel lines



- The indicator shifts the intercept only.
- The effect of the category does not depend on x_1: the vertical gap is β_2 everywhere.
- This is a strong assumption; we will test it with an interaction term.

Tool life: fitted parallel-lines model

$$\hat{y} = 36.986 - 0.0266\,x_1 + 15.004\,x_2.$$

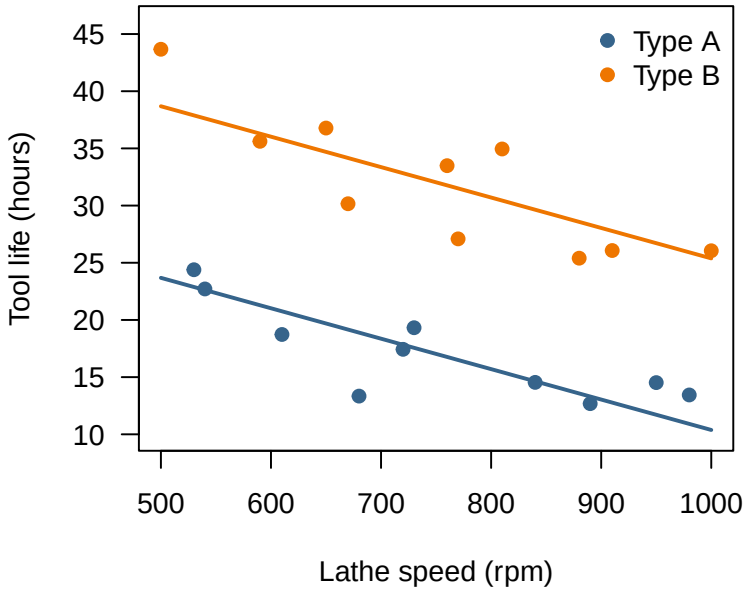
Predictor	Estimate	Std. error	t
Intercept	36.986	3.510	10.54
x_1 (speed)	−0.0266	0.0045	−5.89
x_2 (type B)	15.004	1.360	11.04

$R^2 = 0.900$, $\quad \hat{\sigma} = 3.04$ on 17 degrees of freedom.

Interpreting the coefficients

- **Question:** what do $\hat{\beta}_1 = -0.0266$ and $\hat{\beta}_2 = 15.004$ mean in this problem?

Fitted lines and data



- Type A: $\hat{y} = 36.99 - 0.0266\,x_1$.
- Type B: $\hat{y} = 51.99 - 0.0266\,x_1$.
- Adding the indicator raises R^2 from 0.186 to 0.900 and cuts $\hat{\sigma}$ from 8.44 to 3.04.

3 Inference on the indicator coefficient

Full and reduced models

- A hypothesis H_0 about the coefficients, e.g. that some of them are zero, defines a simpler model.
- **Full model (FM):** all terms; p parameters.
- **Reduced model (RM):** the FM with H_0 imposed; $p - q$ parameters, where q is the number of restrictions in H_0.
- The RM is *nested* in the FM: every RM is a special case of the FM.

Example: $H_0 : \beta_2 = 0$ ($q = 1$).

$$\begin{aligned} \text{FM}: &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon, \\ \text{RM}: &\quad y = \beta_0 + \beta_1 x_1 + \varepsilon. \end{aligned}$$

Extra sum of squares

- Least squares searches over all coefficient values allowed by the model; the FM allows more, so

$$SS_{\text{Res}}(\text{RM}) \geq SS_{\text{Res}}(\text{FM}).$$

- The difference is the **extra sum of squares**: the increase in the residual sum of squares caused by imposing H_0,

$$SS_{\text{Res}}(\text{RM}) - SS_{\text{Res}}(\text{FM}) \geq 0.$$

- Small increase: the dropped terms add little, and H_0 is plausible. Large increase: the dropped terms matter.
- **Question:** how large is "large"?

F-test for nested models

- Under H_0, with n observations, p parameters in the FM and q restrictions:

$$F_0 = \frac{\left[SS_{\text{Res}}(\text{RM}) - SS_{\text{Res}}(\text{FM})\right]/q}{SS_{\text{Res}}(\text{FM})/(n-p)} \sim F_{q,\, n-p}.$$

- Numerator: increase in SS_{Res} per restriction. Denominator: MS_{Res} of the FM, an estimate of σ^2.
- Reject H_0 at level α if $F_0 > F_{\alpha,\, q,\, n-p}$.
- For $q = 1$ the test is equivalent to the two-sided t-test of the coefficient, with $F_0 = t_0^2$.

Inference on the indicator coefficient

- $H_0 \colon \beta_2 = 0$ (the two tool types have the same mean life at every speed) is tested with the usual t-test; here $t_0 = 11.04$ on 17 df.
- Equivalently, use the F-test with FM: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ and RM: $y = \beta_0 + \beta_1 x_1 + \varepsilon$ (speed only); here $q = 1$, $n - p = 17$:

$$F_0 = \frac{[SS_{\text{Res}}(\text{RM}) - SS_{\text{Res}}(\text{FM})]/1}{SS_{\text{Res}}(\text{FM})/17} = \frac{(1282.08 - 157.05)/1}{157.05/17} = 121.8,$$
$$F_0 = t_0^2.$$

- Reject H_0: tool type is strongly associated with tool life after adjusting for speed.

4 Factors with more than two levels

Categorical regressor with k levels

- A factor with k levels is represented by $k - 1$ indicator variables.
- The level without an indicator is the reference level.
- Suppose there were three tool types A, B, C (hypothetical extension), with A as reference:

$$x_2 = \begin{cases} 1 & \text{if type B,} \\ 0 & \text{otherwise,} \end{cases} \qquad x_3 = \begin{cases} 1 & \text{if type C,} \\ 0 & \text{otherwise.} \end{cases}$$

Coding and interpretation for $k = 3$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

Type	x_2	x_3	$\mathbb{E}(y \mid x_1)$
A	0	0	$\beta_0 + \beta_1 x_1$
B	1	0	$(\beta_0 + \beta_2) + \beta_1 x_1$
C	0	1	$(\beta_0 + \beta_3) + \beta_1 x_1$

β_2 and β_3 are the differences of B and C from the reference A, at fixed x_1. The difference between B and C is $\beta_2 - \beta_3$.

The dummy variable trap

- Suppose we include all $k = 3$ indicators $\mathbf{d}_A, \mathbf{d}_B, \mathbf{d}_C$ together with an intercept.
- Every observation belongs to exactly one type, so

$$\mathbf{1} = \mathbf{d}_A + \mathbf{d}_B + \mathbf{d}_C,$$
$$\text{rank}(\mathbf{X}) = 4 < 5 \text{ columns.}$$

- $\mathbf{X}^\top \mathbf{X}$ is singular: the normal equations have no unique solution and β is not identifiable.
- Remedy: drop one indicator (use $k - 1$), or drop the intercept (then each coefficient is a level-specific intercept).

Question: counting indicators

- A factor has 4 levels and the model includes an intercept. How many indicators are needed? How many if the intercept is removed, and how does the meaning of the coefficients change?

Choice of reference level

- Changing the reference level changes the coefficients, not the model.
- With reference C instead of A, the model $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$ is re-expressed as

$$\begin{aligned} \beta'_0 &= \beta_0 + \beta_3, \\ \beta'_A &= -\beta_3, \\ \beta'_B &= \beta_2 - \beta_3. \end{aligned}$$

- Fitted values, residuals, SS_{Res}, R^2 and the overall F-test of the factor are identical for every choice of reference level.
- Choose the reference level that makes the comparisons of interest direct (e.g. a control group).

Testing the factor as a whole

- Individual t-tests compare one level with the reference only.
- To test whether the factor matters at all, $H_0 \colon \beta_2 = \beta_3 = 0$, use the partial F-test:

$$F_0 = \frac{\left[SS_{\text{Res}}(\text{RM}) - SS_{\text{Res}}(\text{FM})\right]/(k-1)}{SS_{\text{Res}}(\text{FM})/(n-p)} \sim F_{k-1,\,n-p} \quad \text{under } H_0,$$

- where the reduced model omits all $k-1$ indicators and p is the number of parameters of the full model.

Indicator variables versus allocated codes

- Tempting alternative: one regressor with codes A = 1, B = 2, C = 3 (allocated codes).
- This forces $\mathbb{E}(y)$ to change by the same amount β from A to B and from B to C, as if the levels were equally spaced on a numeric scale.
- For nominal categories that ordering and spacing is arbitrary; recoding to A = 1, C = 2, B = 3 would change the fit.
- Indicator variables impose no such structure: each level gets its own mean shift.
- Allocated codes are defensible only when the levels are genuinely ordered and roughly equally spaced.

5 Interaction terms

Do both tool types respond to speed in the same way?

- The parallel-lines model assumes that increasing speed shortens life by the same amount for tool types A and B.
- **Question:** how can we let the slope on x_1 differ between the two types?

Indicator $\times$ quantitative interaction

- Add the product of the indicator and the quantitative regressor:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \varepsilon_i.$$

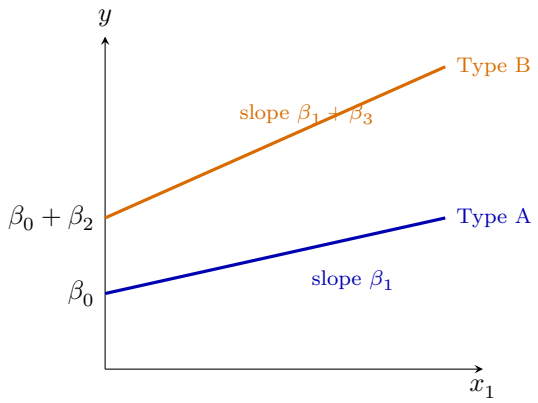
- The effect of x_1 now depends on the category, and the effect of the category depends on x_1.

Group-specific lines

$$\begin{aligned} &\text{Type A } (x_2 = 0): \quad \mathbb{E}(y \mid x_1) = \beta_0 + \beta_1 x_1, \\ &\text{Type B } (x_2 = 1): \quad \mathbb{E}(y \mid x_1) = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\, x_1. \end{aligned}$$

- β_2: difference in intercepts; β_3: difference in slopes.
- The difference between the types at speed x_1 is $\beta_2 + \beta_3 x_1$; it changes linearly with x_1.

Geometry: non-parallel lines



- Two free intercepts and two free slopes.
- The vertical gap between the lines grows with x_1 when $\beta_3 > 0$.
- The lines cross at $x_1 = -\beta_2/\beta_3$ (if this lies in the range of the data, the ranking of the types reverses).

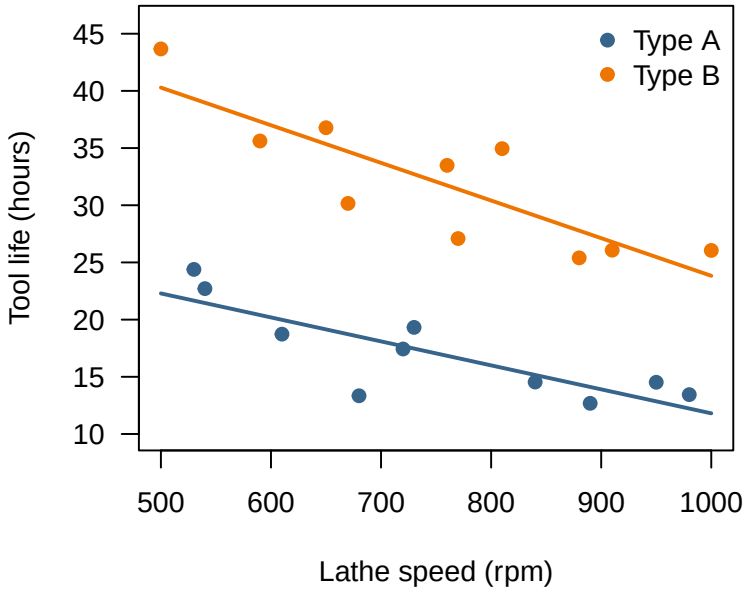
Tool life: fitted interaction model

$$\hat{y} = 32.775 - 0.0210\,x_1 + 23.971\,x_2 - 0.0119\,x_1x_2.$$

Predictor	Estimate	Std. error	t
Intercept	32.775	4.634	7.07
x_1 (speed)	−0.0210	0.0061	−3.45
x_2 (type B)	23.971	6.769	3.54
x_1x_2	−0.0119	0.0088	−1.35

$R^2 = 0.911$, $\quad \hat{\sigma} = 2.97$ on 16 degrees of freedom.

Tool life: different slopes by tool type



- Type A: $\hat{y} = 32.78 - 0.0210\,x_1$.
- Type B: $\hat{y} = 56.75 - 0.0329\,x_1$.
- The estimated slope is steeper for type B ($\hat{\beta}_3 = -0.0119$), so the lines converge as speed increases.
- Is this difference larger than sampling variation?

Testing the interaction

- $H_0 : \beta_3 = 0$ (parallel lines): is the interaction term needed beyond x_1 and x_2? Compare nested models, with $q = 1$ and $n - p = 16$:

$$\begin{aligned} \text{FM:} &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon, \\ \text{RM:} &\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon. \end{aligned}$$

Testing the interaction: result

$$F_0 = \frac{[SS_{\text{Res}}(\text{RM}) - SS_{\text{Res}}(\text{FM})]/1}{SS_{\text{Res}}(\text{FM})/16} = \frac{(157.05 - 140.98)/1}{140.98/16} = 1.82,$$
$$t_0 = -1.35 \quad (t_0^2 = F_0), \qquad p\text{-value} = 0.196.$$

- No evidence that the slopes differ: the parallel-lines model is adequate for these data.
- The estimated slopes differ by 0.0119 per rpm, but with $n = 20$ the estimate is too imprecise to distinguish it from zero.

Testing sequence

- Test the interaction first; effects of the indicator are interpreted in the model that survives.
- Equal lines overall, $H_0 \colon \beta_2 = \beta_3 = 0$ ($q = 2$): FM is the interaction model, RM is the speed-only regression:

$$F_0 = \frac{(1282.08 - 140.98)/2}{140.98/16} = 64.8 \quad (p < 0.001).$$

- Reject: one common line is not adequate; tool type matters.
- Conclusion for tool life: keep x_1 and x_2, drop $x_1 x_2$.

Question: why not fit two separate regressions?

- The interaction model gives one line for each tool type. Why not simply fit two regressions, one for each type?

Indicator variables versus separate regressions

- Separate fits (type A, type B) reproduce the interaction model coefficients:

$$\text{A}\colon \quad \hat\beta_0,\ \hat\beta_1 = 32.775,\ -0.0210,$$
$$\text{B}\colon \quad \hat\beta_0 + \hat\beta_2,\ \hat\beta_1 + \hat\beta_3 = 56.745,\ -0.0329.$$

- The interaction model assumes a common σ^2 and estimates it from all n observations; separate fits estimate one σ^2 per group.
- Only the single model allows direct tests of β_2, β_3, and of the special cases (parallel, equal lines).

Interaction between two quantitative regressors

- Interaction is not limited to indicators:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon.$$

- The effect on $\mathbb{E}(y)$ of a unit change in x_1 depends on the level of x_2:

$$\frac{\partial\, \mathbb{E}(y)}{\partial x_1} = \beta_1 + \beta_{12}\, x_2, \qquad \frac{\partial\, \mathbb{E}(y)}{\partial x_2} = \beta_2 + \beta_{12}\, x_1.$$

- The mean response surface is no longer a plane; slices at fixed x_2 are lines with different slopes (this is the indicator case with $x_2 \in \{0, 1\}$).

Practical guidance on interactions

- With x_1x_2 in the model, β_1 is the effect of x_1 only at $x_2 = 0$; it is not a "main effect" unless $x_2 = 0$ is meaningful.
- Keep the lower-order terms x_1 and x_2 whenever x_1x_2 is included (hierarchy).
- Centering the quantitative regressors at their means makes β_1 and β_2 interpretable at average values and reduces collinearity with x_1x_2.
- Do not interpret interaction coefficients in isolation; plot the fitted lines at chosen values of the other regressor.
- Each additional interaction costs a degree of freedom; add them for substantive reasons, not by default.

Interaction with a k-level factor

- For a three-level factor (indicators x_2, x_3) allow a separate slope for each level:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \varepsilon.$$

- Slopes in x_1: β_1 (reference), $\beta_1 + \beta_4$, $\beta_1 + \beta_5$.
- Test of equal slopes, $H_0 : \beta_4 = \beta_5 = 0$: partial F-test with $k - 1 = 2$ numerator degrees of freedom.

Summary

- An indicator variable encodes a binary category as 0/1; a factor with k levels needs $k - 1$ indicators plus an intercept.
- Indicators shift the intercept: they give parallel lines with common slope. Coefficients are differences from the reference level.
- Including all k indicators with an intercept makes $\mathbf{X}^\top\mathbf{X}$ singular (dummy variable trap).
- The reference level changes coefficients, not the fit; test a whole factor with the partial F-test.
- Interaction terms x_1x_2 let the slope depend on the category (or on another regressor); test them first with the extra-sum-of-squares F-test (RM versus FM).
- Tool life data: type matters ($F_0 = 121.8$); the slope difference is not significant ($p = 0.196$).